



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

UG Programmes CE, CSE, ECE, EEE, IT & ME are Accredited by NBA, Accredited by NAAC with A⁺

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Estd:1980

Regulation: R20			III / IV - B.Tech. I - Semester						
CSE (IoT AND CYBER SECURITY INCLUDING BLOCK CHAIN TECHNOLOGY)									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2022-23 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20AM3101	Design and Analysis of Algorithms	PC	3	3	0	0	30	70	100
B20CI3101	IoT Architecture and its Protocols	PC	3	3	0	0	30	70	100
B20CI3102	Computer Communication and Networks	PC	3	3	0	0	30	70	100
#PE-I	Professional Elective -I	PE	3	2	1	0	30	70	100
#OE-I	Open Elective-I	OE	3	3	0	0	30	70	100
B20CI3107	Network Programming Lab	PC	1.5	0	0	3	15	35	50
B20CI3108	Arduino Lab	PC	1.5	0	0	3	15	35	50
B20CI3109	Web Application Development (Skill Oriented Course)	SOC	2	1	0	2	--	50	50
B20MC3101	Employability Skills	MC	0	3	0	0	--	--	--
B20CI3110	Summer Internship	PR	1.5	--	--	--	--	50	50
TOTAL			21.5	18	1	8	180	520	700

	Course Code	Course
#PE-I	B20CI3103	Compiler design
	B20CI3104	Software Engineering
	B20CI3105	Advanced Computer Architecture
	B20CI3106	Computer Graphics
#OE-I	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B20AM3101	PC	3	0	0	3	30	70	3 Hrs.
DESIGN AND ANALYSIS OF ALGORITHMS								
(Common to AIML, CIC)								
Course Objectives:								
1.	Analyze the asymptotic performance of algorithms							
2.	Write rigorous correctness proofs for algorithms.							
3.	Demonstrate a familiarity with major algorithm design paradigms and data structures							
4.	Synthesize efficient algorithms in common engineering design situations							
Course Outcomes:								
S. No	Outcome							Knowledge Level
1.	Apply mathematical analysis methods to analyse the algorithm running times using asymptotic notations							K4
2.	Compare and understand how the choice of data structures impact the performance of various greedy algorithms							K3
3.	Describe, apply and analyze the complexities of Dynamic Programming Algorithms							K3
4.	Describe, apply and analyze the complexity of Backtracking and Branch and Bound, and explain the situations which call for usage of these paradigms							K3
5.	Infer lower bounds for common problems like searching, sorting, merging, selection, Understand the concepts of P, NP classes							K2
SYLLABUS								
UNIT-I (14Hrs)	INTRODUCTION: Getting Started: Insertion sort, analyzing algorithms, Designing algorithms, Growth of Functions: Asymptotic Notation, Standard notations and common functions, Master method for solving recurrences DIVIDE AND CONQUER: General method, Finding maximum and minimum, Merge sort, Quick sort, Performance Measurement, Selection Problem, A Worst-Case Optimal Algorithm, Strassen's matrix multiplication, convex hull Problem- QuickHull Algorithm Sets and Disjoint set union, Union and Find Operations							
UNIT-II (10 Hrs)	THE GREEDY METHOD: General method, Knapsack problem, Job sequencing with deadlines, Minimum cost spanning trees- Prim's algorithm, Kruskal's algorithm, Optimal storage on tapes, Optimal merge patterns, Huffman coding, Single source shortest paths.							
UNIT-III (12Hrs)	DYNAMIC PROGRAMMING: General method, Multistage graphs, All pairs shortest paths, Optimal binary search trees, String editing, 0/1Knapsack, The travelling salesperson problem							

UNIT-IV (12Hrs)	BACKTRACKING: General method, 8-Queens problem, Sum of subsets, Graph colouring, Hamiltonian cycles BRANCH AND BOUND: The method, Least Cost (LC) Search, The 15-puzzle problem, Control abstractions for LC Search, FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem: LC Branch and Bound, FIFO Branch-and-Bound, Travelling salesperson problem: LC Branch and Bound
UNIT-V (8Hrs)	LOWER BOUND THEORY: The method, Comparison trees, Ordered Searching, Sorting, Selection, Oracles and adversary arguments, Merging, Largest and Second largest NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, Nondeterministic Algorithms, The Classes NP-hard and NP-complete, Cook's theorem. Modular Arithmetic.
Text Books:	
1.	Fundamentals of Computer Algorithms 2nd edition by Ellis Horowitz, Sartaj Sahni, S. Rajasekharan, university press, 2008
2.	Introduction to Algorithms 3rd edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, PHI, 2010
Reference Books:	
1.	Design and Analysis Algorithms – Parag Himanshu Dave, Himanshu Bhalchandra Dave Publisher: Dhanpat Rai & co, 2017
2.	Algorithm Design: Foundations, Analysis and Internet examples, M.T. Goodrich and R. Tomassia, John Wiley and sons, 2006
3.	Introduction to the Design and Analysis of Algorithms, Anany Levitin, PEA, 3rd Edition.
4.	Foundations of Algorithms, R. Neapolitan and K. Naimipour, Jones and Bartlett, Pearson, 2011
5.	Advanced Data Structures – Peter Brass, Cambridge University Press, 2008
e-Resources:	
1.	https://onlinecourses.nptel.ac.in/noc19_cs47/preview
2.	https://onlinecourses.swayam2.ac.in/cec20_cs03/preview
3.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-design-and-analysis-of-algorithms-spring-2015/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3101	PC	3	--	--	3	30	70	3 Hrs.
IoT ARCHITECTURE AND ITS PROTOCOLS								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	Define IoT, describe its impact, and challenges. Explain the IoT architecture and functional stack.							
2.	Review on the sensors and actuators that form the “Things” and appraise the IoT protocols used at different layers.							
3.	Relate Big Data Analytics to IoT and Summarize the protocols for IoT Security.							
4.	Demonstrate a holistic knowledge of IoT application in Oil and Gas Industry and Smart Cities.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the genesis, need, impact, challenges of IoT and IoT architectures							K2
2.	Compare various Sensors, Actuators, Communication Technologies for IoT							K2
3.	Explain the network layer issues, application layer protocols and data analytics for IoT.							K2
4.	Use IoT architecture for Smart Cities.							K3
5.	Use IoT technologies in Oil and Gas, Transportation industries.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to IoT, IoT Network Architecture and Design Genesis of IoT, IoT and Digitization, IoT impact, Connected Roadways; Connected factory, Smart connected buildings, smart creatures; Convergence of IT and OT, IoT Challenges; Drivers Behind Network Architectures, Comparison of IoT architectures; A simplified IoT architecture, The Core IoT functional stack: Layer 1; The Core IoT functional stack: Layer 2; The Core IoT functional stack: Layer 3; IoT Data Management and Compute Sack.							
UNIT-II (10Hrs)	Sensors, Actuators, Communication Technologies Sensors; Actuators, MEMS, Smart Objects; Sensor Networks; Communications Criteria: Range, frequency bands, power consumption; Topology, constrained devices, constrained-node networks; IoT access technologies, IEEE 802.15.4, Zigbee, Zigbee IP; Use cases and topologies of IEEE 802.15.4g, IEEE 802.15.4e, IEEE 1901.2a, IEEE 802.11ah; LoRaWAN, Comparison of various access technologies.							
UNIT-III (10Hrs)	IP as the IoT Network Layer, Application Protocols for IoT Key advantages of Internet Protocol, Adoption or Adaptation of the Internet Protocol, Need for optimization: Constrained nodes, constrained networks; Optimizing IP for IoT: From 6LoWPAN to 6Lo, Header compression; Fragmentation, Mesh addressing, 6Lo							

	working group; RPL; CoAP; Data Analytics for IoT MQTT; Data Analytics for IoT, structured Vs unstructured, data in motion Vs data at rest, IoT Data Analytics overview; IoT Data analytics challenges, getting intelligence from big data; Big data analytics tools and technology: Massively Parallel Processing, NoSQL data bases; Hadoop, Hadoop ecosystem; Edge streaming analytics, comparing big data and edge analytics, edge analytics core function; Distributed analytics system, network analytics.
UNIT-IV (10Hrs)	IoT Security, IoT Case Studies: Smart Cities Common challenges in OT security; The purdue model for control hierarchy, OT network characteristics impacting security; Formal risk analysis structures, OCTAVE and FAIR; The phased application of security in an operational environment; An IoT strategy for smarter cities; Smart city IoT architecture; Smart city security architecture, Smart city use case examples, connected street lighting; Smart parking, smart traffic control, smart environment.
UNIT-V (10Hrs)	IoT Case studies, Oil and Gas industry, Transportation An introduction to Oil and Gas industry; Industry key challenges as digitization drivers; Improving operational efficiency; IoT Architecture for Oil and Gas; Risk control framework for cybersecurity in IoT; IoT use cases for transportation; IoT architecture for transportation: roadways; Connected roadways network architecture.
Textbooks:	
1.	IoT Fundamentals, David Hanes et al. Cisco press/Pearson education. (2018 Indian Reprint)
Reference Books:	
1.	Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015
2.	Internet of Things: Architecture and Design Principles by Raj Kamal, McGraw Hill Education private limited, 2017
3.	Getting Started with the Internet of Things CunoPfister, Oreilly.

Code	Category	T	T	D	C	IM	FM	Exam
R20CTR102	PC	3	--	--	3	30	70	3 Hrs
COMPUTER COMMUNICATION AND NETWORKS								
(For CIC)								
Course Objectives: The main objective of the course is to								
1.	Understand the different types of networks							
2.	Discuss the software and hardware components of a network							
3.	Develop an understanding the principles of computer networks.							
4.	Calculate IPv4 subnet addresses proficiently and explain network layer protocols such as IP, ICMPv4, and routing algorithms							
5.	Explain the functionalities of transport layer protocols (TCP and UDP), including flow control, error control, and congestion control mechanisms. Describe common application layer protocols like HTTP, Telnet, and DNS and their respective roles in networking applications.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain protocol layering, digital, analog signals, data rates and Calculate the performance issues in the physical layer.							K3
2.	Explain transmission media, switching, link layer addressing, and error handling.							K2
3.	Explain various data link layer protocols.							K2
4.	Calculate IPv4 subnet addresses, explain network layer protocols.							K3
5.	Explain transport layer and application layer protocols							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Data communications, Networks, Network Types, Standards and administration; Protocol Layering, TCP/IP Protocol suite, OSI Model (introduction); Introduction to Physical layer: Data and Signals, Periodic analog signals, Digital signals, Transmission impairments, data rate limits, performance.							

UNIT-II (10 Hrs)	The Data Link Layer: Transmission Media: Introduction, Guided media, Un-guided media. Switching: Introduction, Circuit-Switched networks, Packet switching, Structure of a switch. Data Link Layer: Introduction, Link-layer addressing. Error Detection and Correction: Types of errors, Block Coding, Cyclic Redundancy Check, Checksum
UNIT-III (10 Hrs)	Data Link Control: DLC Services, Framing, Finite State Machine (FSM), Stop-and-Wait protocol, HDLC, PPP. Media Access Control (MAC): Random Access, ALOHA, CSMA, CSMA/CD, CSMA/CA, Controlled access: reservation, polling, token passing. Channelization: FDMA, TDMA, CDMA. Introduction to Ethernet and types of Ethernet
UNIT-IV (10 Hrs)	Network Layer: network layer services, packet switching, network layer performance, IPv4 addressing, DHCP, NAT, Forwarding of IP Packets. Network Layer Protocols: Internet Protocol (IP), Datagram Format, ICMPv4, Distance vector and Link state routing. Introduction to IPv6.
UNIT-V (10 Hrs)	Transport Layer: Services, flow control, error control, congestion control, Connection-less and connection-oriented protocols, Stop-and-wait, Go-back-N. UDP and TCP segment formats. TCP services, connection establishment, TCP three-way handshake, TCP States and state transition diagram. Application Layer protocols: HTTP, Telnet, DNS.
Textbooks:	
1.	Behrouz A. Forouzan, Data Communications and Networking, 5 th Edition, McGraw Hill Publication, 2017.
2.	Andrew Tanenbaum, Feamster Wetherall, Computer Networks, 6 th Edition, Global Edition
Reference Books:	
1	James F Kurose, Keith W Ross, "Computer Networking: A Top-Down Approach" 6 th edition
2	Youlu Zheng, Shakil Akthar, "Networks for Computer Scientists and Engineers" Oxford
3	Computer Networks and Internets, Douglas E Comer, fourth Edition, Pearson Education
e-Resources:	
1	https://nntel.ac.in/courses/106105183/25
2	http://www.nntelvideos.in/2012/11/computer-networks.html
3	https://nntel.ac.in/courses/106105183/3

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3103	PE	2	1	0	3	30	70	3Hrs.
COMPILER DESIGN								
(For CIC)								
Pre-requisites: Formal Languages and Automata Theory								
Course Objectives: The main objective of the course is to								
1.	Study the various phases in the design of a compiler.							
2.	Understand the design of top-down, bottom-up parsers and syntax directed translation schemes.							
3.	Introduce LEX and YACC tools.							
4.	Develop algorithms to generate optimized machine code for a target machine.							
Course Outcomes: At the end of the course Students will be able to								
S. No	Outcome							Knowledge Level
1.	Illustrate different phases of compilation process.							K3
2.	Apply Top down and Bottom-Up parsing techniques on CFG to construct parse trees.							K3
3.	Use SDT, Syntax trees to generate intermediate code.							K3
4.	Apply Machine independent optimization techniques.							K3
5.	Apply code generation techniques and Machine dependent Optimization techniques.							K3
SYLLABUS								
UNIT-I (10Hrs)	Language Processors, the structure of a compiler, the science of building a compiler, programming language basics. Lexical Analysis: The Role of the Lexical Analyzer, Input Buffering, Recognition of Tokens, The Lexical-Analyzer Generator Lex, Finite Automata, From Regular Expressions to Automata, Design of a Lexical-Analyzer Generator, Optimization of DFA-Based Pattern Matchers							
UNIT-II (10Hrs)	Syntax Analysis: Introduction, Context-Free Grammars, writing a Grammar, Top-Down Parsing, Recursive and Non-recursive top down parsers, Bottom-Up Parsing, Introduction to LR Parsing: Simple LR, More Powerful LR Parsers, Using Ambiguous Grammars, Parser Generators.							
UNIT-III (10Hrs)	Syntax-Directed Definitions, Evaluation Orders for SDD's, Applications of Syntax-Directed Translation, Syntax-Directed Translation Schemes, and Implementing L-Attributed SDD's. Intermediate-Code Generation: Variants of Syntax Trees, Three-Address Code, Types and Declarations, Type Checking, Control Flow, Back patching, Switch-Statements, Intermediate Code for Procedures.							

UNIT-IV (10Hrs)	Run-Time Environments: Storage organization, Stack Allocation of Space, Access to Nonlocal Data on the Stack, Heap Management, Introduction to Garbage Collection, Introduction to Trace Based Collection. Machine-Independent Optimizations: The Principal Sources of Optimization, Introduction to Data-Flow Analysis, Foundations of Data-Flow Analysis, Constant Propagation, Partial Redundancy Elimination, Loops in Flow Graphs.
UNIT-V (10Hrs)	Code Generation: Issues in the Design of a Code Generator, The Target Language, Addresses in the Target Code, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, A Simple Code Generator. Machine dependent Optimizations: Peephole Optimization, Register Allocation and Assignment, Dynamic Programming Code-Generation.
Text Books:	
1.	Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman, Pearson, Pearson Education India; 2nd edition, 2013
2.	Compiler Construction-Principles and Practice, Kenneth C Loudon, Cengage Learning, 2 nd Edition, 1 January 2011.
Reference Books:	
1.	Modern compiler implementation in C, Andrew W Appel, Revised edition, Cambridge University Press.
2.	The Theory and Practice of Compiler writing, J. P. Tremblay and P. G. Sorenson, TMH
3.	Writing compilers and interpreters, R. Mak, 3rd edition, Wiley student edition.
e-Resources:	
1.	https://nptel.ac.in/courses/106/104/106104123

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3104	PE	2	1	--	3	30	70	3 Hrs.
SOFTWARE ENGINEERING								
(For CIC)								
Course Objectives: The main objectives of the course is to provide students								
1.	Give exposure to phases of Software Development, common process models including Waterfall, the Unified Process, and elements of the Agile Process.							
2.	Give exposure to a variety of Software Engineering practices such as Requirements Analysis and Specification.							
3.	Give exposure to Software Design Techniques.							
4.	Give exposure to various Software Quality Assurance and Testing strategies.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Understand different software process models and their significance.							K2
2.	Distinguish various requirements identification procedures.							K2
3.	Identify different methods for requirement analysis modeling.							K2
4.	Differentiate various aspects of system design and software architectures.							K2
5.	Apply software quality assurance and testing strategies.							K3
SYLLABUS								
UNIT-I (11Hrs)	The Nature of Software, The Unique Nature of WebApps, Software Engineering, The Software Process, Software Engineering Practice, Software Myths. A Generic Process Model, Process Assessment and Improvement, Prescriptive Process Models, Specialized Process Models, The Unified Process, Agility, Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models.							
UNIT-II (08Hrs)	Requirements Engineering, Establishing the Groundwork, Eliciting Requirements, Developing Use Cases, Building the Requirements Model, Negotiating Requirements, Validating Requirements, Requirements Analysis.							
UNIT-III (09 Hrs)	Scenario-Based Modeling, UML Models That Supplement the Use Case, Data Modeling Concepts, Class-Based Modeling, Requirements Modeling Strategies, Flow Oriented							

	Modeling, Creating a Behavioral Model, Patterns for Requirements Modelling, Requirements Modeling for WebApps.
UNIT-IV (13 Hrs)	Design within the Context of Software Engineering, The Design Process, Design Concepts, The Design Model, Software Architecture, Architectural Genres, Architectural Styles, Assessing Alternative Architectural Designs, Architectural Mapping Using Data Flow, Components, Conducting Component-level Design, Component level Design for Web Apps, Designing Traditional Components, Component Based Development. The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, WebApp Interface Design, Design Evaluation.
UNIT-V (09 Hrs)	Elements of Software Quality Assurance, SQA Tasks, Goals & Metrics, Statistical SQA, Software Reliability, A Strategic Approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Test Strategies for Object-Oriented Software, Test Strategies for WebApps, Validation Testing, System Testing, The Art of Debugging, Software Testing Fundamentals, Internal and External Views of Testing, White-Box Testing, Basis Path Testing.
Textbooks:	
1.	Software Engineering a practitioner's approach, Roger S. Pressman, Seventh Edition, McGraw Hill Higher Education.
2.	Software Engineering, Ian Sommerville, Ninth Edition, Pearson.
Reference Books:	
1.	Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010.
2.	Software Engineering, Ugrasen Suman, Cengage
e-Resources:	
1.	https://www.coursera.org/learn/introduction-to-software-engineering
2.	https://onlinecourses.nptel.ac.in/noc21_cs65/preview
3.	https://www.udemy.com/course/complete-software-engineering-course-build-develop-better-software/?couponCode=IND21PM

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3105	PE	2	1	--	3	30	70	3 Hrs.
ADVANCED COMPUTER ARCHITECTURE								
(For CIC)								
Course Objectives: The main objective of the course is to provide students to								
1	Impart the concepts and principles of parallelism.							
2	Develop the designing techniques of Scalable Architectures							
3	Implement the concepts and techniques of advanced computer architectures to design modern computer systems							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Explain types and levels of parallelism.							K2
2	Use pipelining to enhance the system performance							K3
3	Compare VLIW and superscalar processor architectures							K2
4	Illustrate exceptional branching using pipelining							K3
5	Summarize connectivity issues in data parallel architectures.							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction to parallel processing: Basic concepts, Concept of a process program and thread, Processes and threads in languages, Types and Levels of Parallelism: Available and Utilized Parallelism, Types of available parallelism, Levels of available functional Parallelism, utilization of functional Parallelism, Data Parallelism, Classification of Parallelism: Flynn’s Classification, Proposed Classification, Basic Parallel Techniques: Pipelining, Replication.							
UNIT-II (10Hrs)	Introduction to ILP and Pipelined processors: Evolution and Overview of ILP, Dependencies: Data, Control and Resource dependencies. Instruction Scheduling, Speed up potential of ILP, Pipelining: Principles and general structure of pipelines, Performance measures, Application scenarios of pipelines, Pipelined Instruction Processing.							
UNIT-III (10Hrs)	VLIW and Superscalar Processors: Basic Principles, Proposed and Commercial VLIW Architectures, Superscalar Processors: Introduction, parallel decoding, superscalar instruction issue, shelving, register renaming, parallel execution, Preserving sequential consistency of instruction execution.							
UNIT-IV (10Hrs)	Processing of control transfer of instructions: Introduction, Basic approaches to branch handling, delayed branching, branch processing, Multi-way branching Guarded execution.							

UNIT-V (10Hrs)	Data Parallel architecture: Introduction, Connectivity: Near neighbours, Trees and Graphs, The Pyramid, the hypercube, Reconfigurable Networks, Alternative Architectural Classes
Text Books:	
1.	Dezso Sima, Terence Fountain, Peter Kacsuk, Advanced computer architectures: A design space approach, Pearson Education, 1997
2.	Advanced Computer Architecture Second Edition, Kai Hwang, Tata McGraw Hill Publishers.
Reference Books:	
1.	Advanced Computer Architectures, S.G. Shiva, Special Indian edition, CRC, Taylor & Francis.
2.	Computer Architecture, Fourth edition, J. L. Hennessy and D.A. Patterson. ELSEVIER.
3.	Introduction to High Performance Computing for Scientists and Engineers, G. Hager and G. Wellein, CRC Press, Taylor & Francis Group.
4.	Advanced Computer Architecture, D. Sima, T. Fountain, P. Kacsuk, Pearson education.



Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3106	PE	2	1	--	3	30	70	3 Hrs.
COMPUTER GRAPHICS								
(For CIC)								
Course Objectives: The main objective of the course is to provide students to								
1.	Describe the fundamental concepts and theory of computer graphics.							
2.	Apply transformations and modeling computer graphics applications.							
3.	Apply Geometric Transformations.							
4.	Illustrate multimedia authoring techniques and tools.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain the concepts and theory of computer graphics.							K2
2.	Apply 2D Transformations for computer graphics applications.							K3
3.	Apply 2D Viewing techniques for computer graphics applications.							K3
4.	Describe 3-D Object representation.							K2
5.	Apply 3D Transformations for computer graphics applications.							K3
6.	Illustrate multimedia animation techniques and tools.							K2
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to Computer Graphics and Graphic Systems Application areas of computer graphics, overview of graphic systems, video-display devices, Raster-scan systems, Random scan systems, Graphics monitors and work stations and input devices.							
UNIT-II (10 Hrs)	Output Primitives and 2-D Geometrical Transformations Output Primitives: Line drawing algorithms (DDA algorithm, Bresenham's line algorithm), Mid-point circle algorithm, Filled area primitives: Boundary-fill and flood-fill algorithm., Anti aliasing, 2-D Geometrical Transformations: Translation, scaling, rotation, reflection and shear transformation, matrix representations and homogeneous co-ordinates, composite transformations, transformations between coordinates.							

UNIT-III (10 Hrs)	2-D Viewing and Clipping Algorithms: 2-D Viewing: The viewing pipe-line, viewing coordinate reference frame, window to view-port coordinate transformations, Clipping algorithms: Cohen-Sutherland, Liang-Barsky and Nicholl-Lee-Nicholl line clipping algorithms, Sutherland Hodgeman algorithm for clipping polygons.
UNIT-IV (10 Hrs)	3-D Viewing and Object Representation: Spline representation, Hermit curve, Bezier curve and B-spline curve, Polygon surfaces, quadric surfaces, Solid modeling: Techniques for representing solid objects, including wireframe, CSG(Constructive Solid Geometry), B-rep(Boundary Representation). Bezier and B-spline surfaces, Basic illumination models, Shading algorithms, Parallel and Perspective projections.
UNIT-V (10 Hrs)	3D Geometric Transformation and Computer Animation: 3D Geometric Transformation: 3D Translation, rotation, scaling, reflection and shear transformations and composite transformations, Visible surface detection: back-face detection, depth buffer, scan-line method, depth sorting. Computer Animation: Design of animation sequences, general computer animation functions, raster animation, computer animation language, key frame system, motion specification.
Textbooks:	
1.	Hearn, Donald, and M.Pauline Backer.” Computer Graphics C Version Second Edition.”(2021).
2.	Computer Graphics Principles & practice-second edition in C/ Foley, Van Dam, Feiner and Hughes / Pearson Education
Reference Books:	
1.	Computer Graphics Second edition/ Zhigandxiang, Roy Plastock, Schaum’s outlines/Tata Mc-Graw hill edition.
2.	Procedural elements for Computer Graphics/David F Rogers/Tata Mc Graw hill, 2nd edition
3.	Principles of Interactive Computer Graphics/ Neuman and Sproul/TMH.
e-Resources	
1.	Computer Graphics Tutorial - javatpoint
2.	Introduction to Computer Graphics - GeeksforGeeks

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3107	PC	--	--	3	1.5	15	35	3 Hrs.
NETWORK PROGRAMMING LAB								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1	Understand and apply different network commands							
2	Implement different network protocols							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Implement datalink layer framing methods like error control and flow control.							K3
2	Examines and implement the various Routing algorithms							K3
3	Develop client-server applications using sockets.							K3
SYLLABUS								
1	Implement the data link layer framing methods such as character stuffing and bit stuffing.							
2	Write a C program to develop a DNS client server to resolve the given hostname.							
3	Implement on a data set of characters the three CRC polynomials – CRC-12, CRC16andCRC-CCIP							
4	Implement Dijkstra’s algorithm to compute the Shortest path in a graph.							
5	Take an example subnet graph with weights indicating delay between nodes. Now obtain Routing table at each node using distance vector routing algorithm							
6	Take an example subnet of hosts. Obtain broadcast tree for it.							
7	Write a client-server application for chat using TCP							
8	Write a client-server application for chat using UDP							
9	Write a C program to perform sliding window protocol.							
10	Get the MAC or Physical address of the system using Address Resolution Protocol.							
11	Simulate the Implementing Routing Protocols using border gate way protocol(BGP)							
12	Simulate the OPEN SHORTEST PATH FIRST routing protocol based on the cost assigned to the path							
Reference Books:								
1	UNIX Network Programming – Networking APIs: Sockets and XTI by W. Richard Stevens, Volume 1, 2nd Edition, 1998, Prentice Hall.							
2	https://nptel.ac.in/courses/106105183/3							
3	http://vlabs.iitb.ac.in/vlabs-dev/labs_local/computer-networks/labs/explist.php							

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3108	PC	0	0	3	1.5	15	35	3Hrs.
ARDUINO LAB								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	To use Arduino programming and interface various sensor and actuators.							
2.	To design Arduino based IoT applications.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Use sensors, actuators, Arduino in IoT applications.							K3
2.	Analyze various communication protocols for IoT applications.							K4
LIST OF PROGRAMS								
1	Understanding Arduino UNO Board and Components.							
2	Working with Shields.							
3	Blinking LED sketch with Arduino.							
4	Simulation of 4-Way Traffic Light with Arduino.							
5	Using Pulse Width Modulation.							
6	LED Fade Sketch and Button Sketch.							
7	Analog Input Sketch (Bar Graph with LEDs and Potentiometre).							
8	Digital Read Serial Sketch (Working with DHT/IR/Gas or Any other Sensor)							
9	Working with Adafruit Libraries in Arduino.							
10	Spinning a DC Motor and Motor Speed Control Sketch							
11	Interfacing Arduino with Cloud (Thingspeak API).							
ADDITIONAL PROGRAMS								
1	Demonstrate controlling devices through ThingSpeak Cloud using ESP32 WiFi Module.							
2	Interfacing display units like Seven segment and LCD to Arduino.							
3	Interfacing Ultrasonic sensor for distance measurement.							
4	Interfacing PIR Motion sensor.							
Text Books:								
1.	Internet of Things: Architecture, Design Principles and Applications, Rajkamal, McGraw Hill Higher Education. 2017							
2.	Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madhetti, Universities Press, 1st edition, 2014.							
Reference Books :								
1.	Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley, 1st edition,							

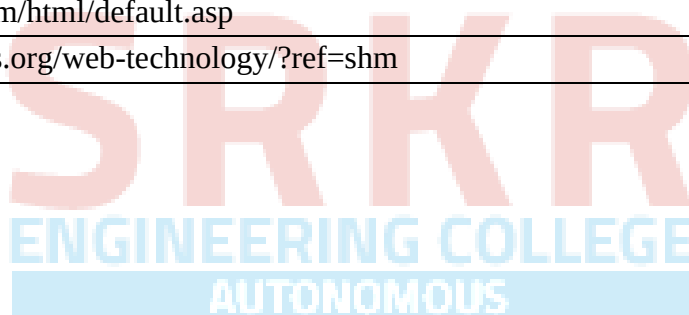
	2014.
2.	1. Getting Started with the Internet of Things CunoPfister,Oreilly. 2011.
3.	Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD),2014.
e-Resources:	
1.	Introduction to Internet of Things, https://swayam.gov.in/nd1_noc20_cs66/previe0077
2.	An Introduction to Programming the Internet of Things(IoT) specialization, https://www.coursera.org/specializations/iot



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3109	SOC	1	--	2	2	--	50	3 Hrs.
WEB APPLICATION DEVELOPMENT								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1	Provide understanding about the core concepts of frontend programming for web application.							
2	Build strong foundation of JavaScript which will help developer to apply JavaScript concepts for responsive web frontend development							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Use HTML to implement static web pages							K3
2	Apply CSS and bootstrap for implementing web pages							K3
3	Use Java Script to implement dynamic web pages							K3
SYLLABUS								
1	HTML ● Introduction to HTML ● Tags, Attribute and Elements, Comments ● Headings, Paragraphs, and Formatting Text ● Lists and Links ● Images and Tables ● Forms and Form Elements							
2	CSS & Bootstrap ● Introduction CSS ● Applying CSS to HTML ● Selectors, Properties and Values ● CSS Colors and Backgrounds, CSS Box Model ● CSS Margins, Padding, and Borders ● CSS Text and Font Properties ● Grid System in Bootstrap ● Buttons, Tables ● Vertical Forms, Horizontal Forms, Inline Forms ● Dropdowns and Responsive Tabs							

3	JavaScript • Introduction to JavaScript • Applying JavaScript (internal and external) • Introduction to Document and Window Object • Variables and Operators • Data Types and Num Type Conversion • Math and String Manipulation • Objects and Arrays , Date and Time • Conditional Statements, Looping in JS • Functions and Event Handling
Reference Books:	
1	Programming the World Wide Web, 7th Edition Robert W Sebesta, Pearson, 2013.
2	WebTechnologies,1st Edition 7th impression, Uttam K Roy, Oxford, 2012.
3	Mean Stack Development,1st Edition, ELadElrom, A press O'Reilly,2016
4	Script & jQuery the missing manual, 2nd Edition, David sawyer mcfarland, O'Reilly, 2011.
5	Java 5. Web Hosting for Dummies, 1st Edition, Peter Pollock, John Wiley & Sons, 2013.
6	RESTful web services, 1st Edition, Leonard Richardson, Ruby, O'Reilly, 2007.
Online Resources:	
1	https://www.w3schools.com/html/default.asp
2	https://www.geeksforgeeks.org/web-technology/?ref=shm



Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC3101	MC	3	--	--	--	--	--	3 Hrs.
EMPLOYABILITY SKILLS								
(Common to CIC, Civil, EEE & Mech)								
Part-A: Verbal Ability								
Course Objectives:								
1.	To introduce concepts required in framing grammatically correct sentences and identifying errors While using Standard English.							
2.	To familiarize the learner with high frequency words as they would be used in their professional career.							
3.	To inculcate logical thinking in order to frame and use data as per the requirement							
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.							
5.	To familiarize students with soft skills and how it influences their professional grow.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/company specific tests and frame grammatically Correct sentences while writing.							K3
2	Answer questions on synonyms, antonyms and other vocabulary-based Exercises while attempting CAT, GRE, GATE and other related tests.							K3
3	Use their logical thinking ability and solve questions related to analogy, Syllogisms, and other reasoning-based exercises.							K3
4	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.							K3
SYLLABUS								
UNIT-I	Spotting Errors, Sentence Improvement							
UNIT-II	Synonyms, Antonyms, Frequently Confused Words, Foreign Phrases, Idioms and Phrasal Verbs, Collocations.							
UNIT-III	Foreign Phrases, Idioms and Phrasal Verbs, Collocations, Analogies, Odd One Out							
UNIT-IV	Sentence completion, Sentence Equivalence, Close Test							
UNIT-V	Reading Comprehension, Para Jumbles							
Text Books:								
1.	Oxford Learners,, Grammar–Finder by John Eastwood, Oxford Publication.							

2.	RS Agarwal books on objective English and verbal reasoning	
3.	English Vocabulary in Use-Advanced, Cambridge University Press	
4.	Collocations In Use, Cambridge University Press	
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt .Ltd.	
6.	Soft Skills, by Dr.K.Alex, S. Chand & Company Ltd., New Delhi	
Reference Books:		
1.	English Grammar in Use by Raymond Murphy, CUP	
2.	Websites: Indiabix,800score, official CAT, GRE and GMAT sites	
3.	Material from IMS, Career Launcher and Time institutes for competitive exams	
4.	The Art of Public Speaking by Dale Carnegie	
5.	The Leader in You by Dale Carnegie	
6.	Emotional Intelligence by Daniel Golman	
7.	Stay Hungry Stay Foolish by Rashmi Bansal	
8.	I have a Dream by Rashmi Bansal.	
Part-B: Quantitative Aptitude-I		
Course Objectives:		
1.	To familiarize students with basic problems on numbers and ratios problems.	
2.	To enrich the skills of solving problems on time, work, speed, distance and also Measurement of units.	
3.	To enable the students to work efficiently on percentage values related to shares, profit and Loss problems.	
4.	To inculcate logical thinking by exposing the students to reasoning related questions.	
5.	To inculcate logical thinking by exposing the students to reasoning related questions.	
Course Outcomes:		
S.No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	The students will be able to solve problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of Syllogisms and be able to effectively attempt competitive	K3

	examinations like CAT, GRE, GATE for further studies	
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three Different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on manpower and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, market price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	
UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.	
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.	
Text Books:		
1.	Quantitative aptitude by RS Agarwal	
2.	Verbal and nonverbal reasoning by RS Agarwal	
3.	Puzzles to puzzle you by shakunatala devi.	
References:		
1.	Barrons by Sharon Welner Green and IraK Wolf (Galgotia Publications pvt. Ltd.)	
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites	
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams	

4.	Books for CAT by Arun sharma.
5.	Elementary and Higher algebra by HS Hall and SR Knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site



SRKR
ENGINEERING COLLEGE
AUTONOMOUS



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

UG Programmes CE, CSE, ECE, EEE, IT & ME are Accredited by NBA, Accredited by NAAC with A⁺

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

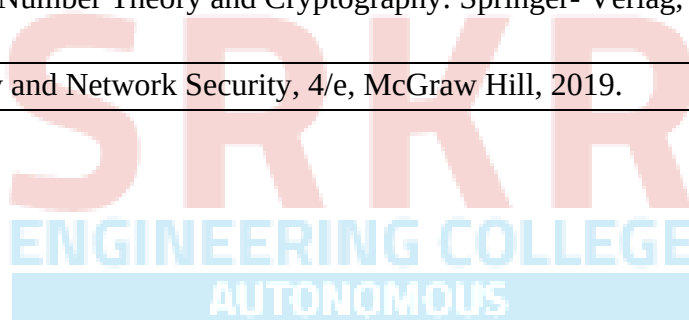
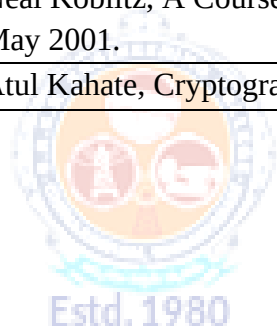
Regulation: R20		III / IV - B.Tech. II - Semester							
CSE (IoT AND CYBER SECURITY INCLUDING BLOCK CHAIN TECHNOLOGY)									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2022-23 admitted Batch onwards)									
Course Code	Course Name	Category	Cr	L	T	P	Int. Marks	Ext. Marks	Total Marks
B20CI3201	Cryptography and Network Security	PC	3	3	0	0	30	70	100
B20CI3202	Introduction to Cyber Security	PC	3	3	0	0	30	70	100
B20CI3203	Block chain technologies	PC	3	3	0	0	30	70	100
#PE-II	Professional Elective -II	PE	3	2	1	0	30	70	100
#OE-II	Open Elective-II	OE	3	3	0	0	30	70	100
B20CI3208	Cyber Security Lab	PC	1.5	0	0	3	15	35	50
B20CI3209	IoT Lab	PC	1.5	0	0	3	15	35	50
B20CI3210	Cryptography and Network Security Lab	PC	1.5	0	0	3	15	35	50
B20HS3203	Soft Skills (Skill Oriented Course)	SOC	2	1	0	2	--	50	50
B20MC3202	Competitive programming	MC	0	0	0	3	--	--	--
B20HS3204	*Gender Sensitization	HS	0	2	0	0	--	--	--
TOTAL			21.5	17	1	14	195	505	700

	Course Code	Course
#PE-II	B20CI3204	Data warehousing & Data Mining
	B20CI3205	Object Oriented Analysis & Design Using UML
	B20CI3206	NoSQL Databases
	B20CI3207	MOOCs-I(NPTEL/SWAYAM)
#OE-II	Student has to study one Open Elective offered by CE or ECE or EEE or ME or S&H from the list enclosed.	

***Note:** Gender Sensitization is a Self-Learning noncredit Audit Course

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3201	PC	3	0	0	3	30	70	3Hrs.
CRYPTOGRAPHY AND NETWORK SECURITY								
(For CIC)								
Pre-requisites: Computer Networks, Data Communications								
Course Objectives: The main objective of the course is to provide students								
1.	Overview of computer security and classical encryption techniques.							
2.	Working principles and utilities of various cryptographic algorithms including symmetric key cryptography and public key cryptography algorithms.							
3.	Design issues and working principles of hashing, message digest algorithms and various authentication protocols.							
4.	Various secure communication protocols standards and Concepts of firewalls.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Illustrate various types of Security attacks and Block Cipher design principles.							K2
2.	Apply Symmetric Key Cryptography techniques to achieve confidentiality service and concepts of Number Theory.							K3
3.	Apply Public Key Cryptography Algorithms to achieve confidentiality and authentication service.							K3
4.	Use Hashing algorithms to achieve Integrity service.							K3
5.	Describe User Authentication and Email, IP security protocols.							K2
6.	Explain Transport level security protocols and the Importance of system security through firewalls.							K2
SYLLABUS								
UNIT-I (8 Hrs)	Classical Encryption Techniques: Security Attacks, Services & Mechanisms, Symmetric Cipher Model. Phishing Attack, Web Based Attacks, SQL Injection Attacks, TCP session hijacking, UDP Session Hijacking. Block Ciphers: Traditional Block Cipher Structure, Block Cipher Design Principles.							
UNIT-II (10 Hrs)	Symmetric Key Cryptography: Data Encryption Standard (DES), Advanced Encryption Standard (AES), Blowfish, IDEA, Block Cipher Modes of Operations. Number Theory: Prime and Relatively Prime Numbers, Modular Arithmetic, Fermat's and Euler's Theorems, The Chinese Remainder Theorem, Discrete Logarithms.							
UNIT-III (12 Hrs)	Public Key Cryptography: Principles, Public Key Cryptography Algorithms, RSA Algorithm, Diffie Hellman Key Exchange, Elliptic Curve Cryptography. Cryptographic Hash Functions: Application of Cryptographic Hash Functions, Requirements & Security, MD5, SHA-512, Message Authentication Functions, Requirements & Security, HMAC & CMAC.							

UNIT-IV (10 Hrs)	Digital Signatures: Digital Signature Algorithm, Key Management and Distribution. User Authentication: Remote User Authentication Principles, Kerberos. Electronic Mail Security: Pretty Good Privacy (PGP) And S/MIME. IP Security: IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload and Key Management: OAKLEY,ISAKMP .
UNIT-V (10Hrs)	Transport Level Security: Web Security Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS) Firewalls: Characteristics, Types of Firewalls, Placement of Firewalls, Firewall Configuration, Trusted Systems.
Text Books:	
1.	Cryptography and Network Security – Principles and Practice, William Stallings, 7/e. Pearson Education, 2017.
2.	Cryptography and Network Security- Behrouz A Forouzan, Debdeep Mukhopadhyaya, McGrawHill, 3rd Edition, 2015.
Reference Books:	
1.	Neal Koblitz, A Course in Number Theory and Cryptography: Springer- Verlag, New York Inc. May 2001.
2.	Atul Kahate, Cryptography and Network Security, 4/e, McGraw Hill, 2019.



Code	Category	L	T	P	C	IM	EM	Exam
R20CI3202	DC	3	--	--	3	30	70	3 Hrs
INTRODUCTION TO CYBER SECURITY								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	Assess the vulnerability of and apply basic protections for the network, operating systems, and endpoints. To maintain the integrity, confidentiality, and availability of your network and your data							
2.	Learn about different ways to monitor your network and how to evaluate alerts that you receive.							
3.	Delve into tools and techniques used to protect your network, including access control, firewalls, cloud security.							
4.	Create policy documents that cover governance and compliance, complying with standards of ethics and legal and regulatory frameworks.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1.	Explain various threats, mitigate common attacks, apply security services.							K2
2.	Use a layered defense-in-depth cybersecurity strategy.							K3
3.	Summarize wireless, mobile, IoT security issues, along with usage and operation of AAA services.							K2
4.	Determine cloud and data security technologies, evaluate alerts, create policy documents that cover governance and compliance. Test and assess network security							K3
5.	Explain network and systems vulnerabilities, risk management plan, forensic investigation, disaster recovery methods.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Cyber Treats and Attacks Types of cyber threats, Deception (Social Engineering) and defending against deception, Types of Cyber Attacks, Wireless and Mobile device attacks, Types of application attacks; Securing Networks: Current state and who is attacking (Threat actors and Cyber criminals), IP Vulnerabilities, ICMP attacks, TCP and UDP Vulnerabilities and attacks; ARP attacks, DNS attacks, DHCP attacks; Common HTTP Exploits, SQL Injection, Cross Site Scripting; Mitigating common network attacks; Fundamentals of Wireless LAN,							

	Types of Wireless LAN Threats; Securing Wireless LAN; Network Security Infrastructure Devices; Security services.
UNIT-II (10 Hrs)	Defense-in-depth Common Windows Operating System Vulnerabilities, Windows Architecture; Windows configuration and monitoring, Windows Security; Operating Systems Security: Patch Management, End point Security, Host Encryption, Boot Integrity; Apple security features, Physical protection for devices. End point Threats, End point security. Host based Malware protection; Network based Malware protection.; Host based firewalls; Host based Intrusion Detection, Application Security; The Cyber Security cube, CIA Triad, Measure to ensure availability, States of data.; Hardware based and software based Cyber security countermeasures. Policies, standards, guidelines, and procedures.; Defense-in-depth: Asset, Vulnerabilities, and Threats. Cyber Security Operations Management. Business Policy, Security Policy, BYOD policy.; Physical security methods, Application Security.; Network Hardening: Services and protocols, Network Hardening: Segmentation.
UNIT-III (10 Hrs)	AAA Wireless device security, Mobile device security, Various measures for Cyber security resilience.; Threats on Embedded and IoT systems, Access Control types, Definitions: Identity management, Authentication, Authorization; Zero Trust Security, Access Control Methods, Network Access Control (NAC) Systems, Account Management; AAA Usage and Operation; Introduction to Access Control Lists, Wild card masking, Configure ACLs.; Implementing ACLs; Mitigating Attacks with ACLs, IPv6 ACLs; Secure Networks with Firewalls, Firewalls in Network Design; Zone-Based Policy Firewalls (ZPF) Overview, ZPF Operation; ZPF Configuration.
UNIT-IV (10 Hrs)	Cloud, IT Security and Governance Virtualization and Cloud Computing, Domains of Cloud Security, Cloud Infrastructure Security, Cloud Application Security; Cloud Data Security, Protecting Virtual Machines; Security Monitoring Protocols, Security Technologies; Security Data: Types of security data, End Device Logs, Network Logs; Evaluating Alerts: Sources of alerts, Alert Evaluation; Governance: Definition, Key roles, Cyber Security policies, Types of Security Policies.; Ethics: Ethics of Cyber Security Specialist, Ten Commandments of Computer Ethics, Types of Cybercrime, Various types of cyber laws.; IT Security Management Framework: Twelve Domains of Cybersecurity, Introduction to ISO 27000, National Cybersecurity Workforce Framework, CIS Critical Security Controls and CCM.; Network Security Assessment, Network Security Testing Techniques, Network Security Testing Tools.; Introduction to Penetration Testing.
UNIT-V (10 Hrs)	CVSS, Risk, Incident Response: Threat Intelligence Sources and Services, Network and Server Profiling, Network

	Anomaly Detection, Network Vulnerability Testing.; Common Vulnerability Scoring System (CVSS); Secure Device Management; Risk Management and Risk Assessment; Security Controls; Evidence Handling and Attack Attribution; The Cyber Kill Chain; The Diamond Model of Intrusion Analysis; Incident Response; Disaster Recovery.
Textbooks:	
1.	Cybersecurity Essentials, Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short, Sybex, O'Reilly Publications.
Reference Books:	
1.	Cyber Security Essentials. Cisco Networking Academy, Skills for All.



Code	Category	L	T	P	C	IM	EM	Exam
R20C13203	PC	3	--	--	3	30	70	3 Hrs
BLOCKCHAIN TECHNOLOGIES								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	Understand how block chain systems (mainly Bit coin and Ethereum) work and to securely interact with them.							
2	Design, build, and deploy smart contracts and distributed applications							
3	Integrate ideas from block chain technology into their own projects							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1	Explain the fundamental concepts of the Block chain technology							K1
2.	Summarize Block chain concepts and the risks involved in building its applications.							K2
3	Determine various Block chain Solutions for designing Applications							K3
4	Analyze concepts of Ethereum for implementing Block Chain							K3
5	Use Hyper ledger to solve different use cases							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction, Scenarios, Challenges Articulated Block chain, Block chain Characteristics, Opportunities Using Block chain, History of Block chain, Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Block chain Evolution, Consortia, Forks, Public Block chain Environments, Type of Players in Block chain Ecosystem, Players in Market.							
UNIT-II (10 Hrs)	Block chain Concepts: Introduction, Changing of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on block chain data storage on block chain, wallets, coding on block chain: smart contracts, peer-to-peer network, types of block chain nodes, risk associated with block chain solutions, life cycle of block chain transaction.							

UNIT-III (10 Hrs)	Architecting Block chain solutions: Introduction, Obstacles for Use of Block chain, Block chain Relevance Evaluation Framework, Block chain Solutions Reference Architecture, Types of Block chain Applications, Cryptographic Tokens, Typical Solution Architecture for Enterprise Use Cases, Types of Block chain Solutions, Architecture Considerations, Architecture with Block chain Platforms, Approach for Designing Block chain Applications.
UNIT-IV (10 Hrs)	Ethereum Block chain Implementation: Introduction, Tuna Fish Tracking Use Case, Ethereum Ecosystem, Ethereum Development, Ethereum Tool Stack, Ethereum Virtual Machine, Smart Contract Programming, Integrated Development Environment, Truffle Framework, Ganache, Unit Testing, Ethereum Accounts, My Ether Wallet, Ethereum Networks/Environments, Infura, Ether scan, Ethereum Clients, Decentralized Application, Meta mask. Tuna Fish Use Case Implementation, Open Zeppelin Contracts
UNIT-V (10 Hrs)	Hyper ledger Block chain Implementation, Introduction, Use Case – Car Ownership Tracking, Hyper ledger Fabric, Hyper ledger Fabric Transaction Flow, Fab Car Use Case Implementation, Invoking Chain, code Functions Using Client Application., Advanced Concepts in Block chain: Introduction, Inter Planetary File System (IPFS), Zero-Knowledge, Proofs, Oracles, Self-Sovereign Identity, Block chain with IoT and AI/ML Quantum Computing and Block chain, Initial Coin Offering, Block chain Cloud Offerings, Block chain and its Future Potential.
Textbooks: Estd. 1980 AUTONOMOUS	
1.	Ambadas, Arshad Sarfarz Ariff, Sham “Block chain for Enterprise Application Developers”, Wiley
2.	Andreas M. Antonopoulos, “Mastering Bitcoin: Programming the Open Block chain” , O’Reilly
Reference Books:	
1.	Block chain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, Mc Graw Hill.
2.	Block chain: Blueprint for a New Economy, Melanie Swan, O’Reilly
e-Resources	
1	https://www.coursera.org/specializations/blockchain
2	https://www.coursera.org/learn/blockchain-basics
3	https://onlinecourses.nptel.ac.in/noc22_cs44/preview

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3204	PE	2	1	0	3	30	70	3Hrs.
DATA WAREHOUSING & DATA MINING								
(For CIC)								
Pre-requisites: Data Structures, Basics of Probability and Statistics like Prior and posterior Probabilities, Bayes Theorem, Mean, Median, Mode, Standard Deviation etc.								
Course Objectives: The main objective of the course is to								
1.	Introduce fundamental concepts, principles, and techniques of data warehousing and data mining.							
2.	Provide a comprehensive understanding of data preprocessing techniques and their importance in data mining.							
3.	Familiarize with association analysis and its applications in discovering interesting patterns in large datasets.							
4.	Equip with the knowledge and skills necessary for classification tasks, including decision tree induction, naive Bayes classification, and model evaluation.							
5.	Introduce cluster analysis and anomaly/outlier detection methods and their practical applications in various domains.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Illustrate multidimensional data model, data warehouse design, implementation and concepts of data mining.							K2
2.	Demonstrate data preprocessing techniques and Association Rule Mining Algorithms.							K3
3.	Apply classification algorithms like decision tree induction, Bayes Theorem, KNN to solve classification problems.							K3
4.	Demonstrate the performance of classifier using holdout method, random subsampling, cross-validation, bootstrap.							K3
5.	Use clustering techniques inclusive of k-means, k-medoids, agglomerative hierarchical clustering, and DBSCAN to analyze their strengths and weakness.							K3
6.	Describe the Characteristics and various approaches for anomaly detection.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Data Warehouse: Basic Concepts, A Multidimensional Data Model, Data Warehouse Architecture, Data Warehouse Design and Usage, Data Warehouse Implementation. (Textbook 2)							
	Data Mining: Introduction, What is Data Mining? Motivating challenges, The origins of Data Mining, Data Mining Tasks, Types of Data, Data Quality. (Textbook 1)							

UNIT-II (10 Hrs)	Data Preprocessing: Aggregation, Sampling, Feature Subset Selection, Feature Creation, Discretization and Binarization, Variable Transformation, Measures of Similarity and Dissimilarity. (Textbook 1) Association Analysis: Basic Concepts and Algorithms: Problem Definition, Frequent Item Set Generation, Apriori Principle, Apriori Algorithm, Rule Generation, FP-Growth Algorithm. (Textbook 1)
UNIT-III (10 Hrs)	Classification: Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Algorithm, Attribute Selection Measures, Bayes Theorem, Naïve Bayes Classifier, KNN (Textbook 2) Model Overfitting: Due to the presence of noise, due to lack of representation samples, evaluating the performance of classifier: hold out method, random subsampling, cross-validation, bootstrap. (Textbook 1)
UNIT-IV (10 Hrs)	Cluster Analysis: Basic Concepts and Algorithms: Overview, What Is Cluster Analysis? Different Types of Clustering, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bisecting K-means, Strengths and Weaknesses; K-medoids, Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses. (Textbook 1)
UNIT-V (10 Hrs)	Anomaly / Outlier Detection: Characteristics of anomaly detection problem, Characteristics of anomaly detection methods, Statistical approaches, Proximity based approaches, Clustering based approaches, Information theoretic approaches, Evaluation of Anomaly Detection (Textbook 1)
Text Books:	
1.	Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Second Edition, Pearson, 2019.
2.	Data Mining Concepts and Techniques, 3rd Edition, Jiawei Han, Michel Kamber, Elsevier, 2011
Reference Books:	
1.	Data Mining Techniques and Applications: An Introduction, Hongbo Du, Cengage Learning, 2010
2.	Data Mining: Introductory and Advanced Topics: Dunham, First Edition, Pearson, 2020
3.	Data Warehousing, Data Mining & OLAP, Alex Berson, Stephen Smith, McGraw Hill Education, 2017
4.	Data Mining Techniques, Arun K Pujari, Universities Press, 2001
e-Resources:	
1.	NPTEL Online Course on Data Mining: https://onlinecourses.nptel.ac.in/noc21_cs06/preview

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3205	PE	2	1	0	3	30	70	3Hrs.
OBJECT ORIENTED ANALYSIS & DESIGN USING UML								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	Become familiar with all phases of OOAD.							
2.	Learn the Object design Principles and understand how to apply them towards Implementation.							
3.	Master the main features of the UML.							
4.	Master the main concepts of Object Technologies and how to apply them at work and develop the ability to analyze and solve challenging problems in various domains.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Summarize the nature of a complex system and its solutions.							K2
2.	Apply object-oriented concepts to identify classes, objects and relationships among them.							K3
3.	Explain the conceptual and structural modeling techniques of UML							K2
4.	Model basic behavior of a software system with Use Case, Interaction and Activity Diagrams.							K3
5.	Model advanced behavioral aspects and Runtime environment of Software Systems.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: The Structure of Complex systems, The Inherent Complexity of Software, Attributes of Complex System, Organized and Disorganized Complexity, Bringing Order to Chaos, Designing Complex Systems, Evolution of Object Model, Foundation of Object Model, Elements of Object Model, Applying the Object Model.							
UNIT-II (10 Hrs)	Classes and Objects: Nature of object, Relationships among objects, Nature of a Class, Relationship among Classes, Interplay of Classes and Objects, Identifying Classes and Objects, Importance of Proper Classification, Identifying Classes and Objects, Key abstractions and Mechanisms.							
UNIT-III (10 Hrs)	Introduction to UML: Why we model, Conceptual model of UML, Architecture, Classes, Relationships, Common Mechanisms, Class diagrams, Object diagrams.							
UNIT-IV (10 Hrs)	Basic Behavioral Modeling: Interactions, Interaction diagrams, Use cases, Use case Diagrams, Activity Diagrams.							

UNIT-V (10 Hrs)	Advanced Behavioral Modeling: Events and signals, state machines, processes and Threads, time and space, state chart diagrams. Architectural Modelling: Component, Deployment, Component diagrams and Deployment diagrams. Case Study: The Unified Library application.
Text Books:	
1.	“Object- Oriented Analysis And Design with Applications”, Grady BOOCH, Robert A. Maksimchuk, Michael W. ENGLE, Bobbi J. Young, Jim Conallen, Kellia Houston, 3rd edition, 2013, PEARSON.
2.	“The Unified Modeling Language User Guide”, Grady Booch, James Rumbaugh, Ivar Jacobson, 12th Impression, 2012, PEARSON.
Reference Books:	
1.	“Object-oriented analysis and design using UML”, Mahesh P. Matha, PHI
2.	“Head first object-oriented analysis and design”, Brett D. McLaughlin, Gary Pollice, Dave West, O’Reilly
3.	“Object-oriented analysis and design with the Unified process”, John W. Satzinger, Robert B. Jackson, Stephen D. Burd, Cengage Learning Jacobson, Grady Booch, Addison-Wesley



Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3206	PE	2	1	0	3	30	70	3Hrs.
NoSQL DATABASES								
(For CIC)								
Course Objectives: The main objective of the course is to provide students								
1.	Define, compare and use the four types of NoSQL Databases (Document-oriented, Key-Value Pairs, Column-oriented and Graph).							
2.	Know the CRUD operation in NoSQL Databases.							
3.	Learn Storage architecture of NoSQL Databases.							
4.	Know Accessing Data and schema evolution in column-oriented Databases and Key/Value stores.							
5.	Learn creating and using indexes in NoSQL databases.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Identify what type of NoSQL database to implement based on business requirements (key-value, document, full text, graph, etc.)							K2
2.	Apply CRUD operations in NoSQL Databases.							K3
3.	Understand Storage architecture of NoSQL Databases							K2
4.	Demonstrate Accessing Data and schema evolution in column-oriented Databases and Key/Value stores.							K3
5.	Explain creating and using indexes in NoSQL databases like Mongoddb, Couchdb and Apache Cassandra.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to NoSQL: Definition And Introduction, Sorted Ordered Column-Oriented Stores, Key/Value Stores, Document Databases, Graph Databases, Examining Two Simple Examples, Location Preferences Store, Car Make And Model Database, Working With Language Bindings.							
UNIT-II (10 Hrs)	Interacting with NoSQL: If NoSql Then What, Language Bindings For NoSQL Data Stores, Performing Crud Operations, Creating Records, Accessing Data, Updating And Deleting Data.							
UNIT-III (10 Hrs)	NoSQL Storage Architecture: Working With Column-Oriented Databases, Hbase Distributed Storage Architecture, Document Store Internals, Understanding Key/Value Stores In Memcached And Redis, Eventually Consistent Non-Relational Databases.							

UNIT-IV (10 Hrs)	NoSQL Stores: Similarities Between Sql And Mongodb Query Features, Accessing Data From Column-Oriented Databases Like Hbase, Querying Redis Data Stores, Changing Document Databases, Schema Evolution In Column-Oriented Databases, Hbase Data Import And Export, Data Evolution In Key/Value Stores.
UNIT-V (10 Hrs)	Indexing and Ordering Data Sets : Essential Concepts Behind A Database Index, Indexing And Ordering In Mongodb, Creating and Using Indexes In Mongodb, Indexing And Ordering In Couchdb, Indexing In Apache Cassandra.
Text Books:	
1.	Pramod Sadalage and Martin Fowler, NoSQL Distilled, Addison-Wesley Professional, 2012.
2.	Dan McCreary and Ann Kelly, Making Sense of NoSQL, Manning Publications, 2013.
Reference Books:	
1.	Shashank Tiwari, Professional NoSQL, Wrox Press, Wiley, 2011, ISBN: 978-0-470- 94224-6
2.	Gaurav Vaish, Getting Started with NoSQL, Packt Publishing, 2013.



Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3207	PE	2	1	--	3	30	70	3 Hrs.
MOOCS-I								
(For CIC)								
MOOCS-I course should belong to the B.Tech. Programme and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCS will be mapped to the grade table given in the Academic Regulations.								



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3208	PC	--	--	3	1.5	15	35	3 Hrs.
CYBER SECURITY LAB								
(For CIC)								
Course Objectives: The main objective of the course is to provide students to								
1	Investigate threat landscape, explore and configure windows host hardening methods.							
2	Explore and configure Linux security issues and use methods to harden Linux machine.							
3	Implement AAA on Packet Tracer.							
4	Apply IDS and Firewall rules.							
5	Analyze attacks and risk.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1	Perform database attacks, analyze DNS traffic, hardening of windows host for security.							K3
2	Configure Linux machine to harden the security.							K3
3	Implement AAA services using Packet Tracer simulation tool.							K3
4	Configure firewall and IDS rules.							K3
5	Analyse risks and attacks.							K4
SYLLABUS								
1	1. Investigate Threat Landscape using Packet Tracer. 2. Exploring DNS Traffic 3. Install a Virtual Machine on a Personal Computer. 4. Attacking a mySQL Database.							
2	1. Reading Server Logs. 2. Configure basic wireless security, Trouble shoot wireless connectivity. 3. Packet Tracer - Access Control List Demonstration 4. Exploring Processes, Threads, Handles, and Windows Registry.							
3	1. Create Windows User accounts, Using Windows Power Shell, Windows Task Manager, Monitor and Manage system resources in Windows. 2. Linux Port scanning using nmap. Linux utilities ps, netstat, telnet commands. 3. Linux: locating log files, file permissions, Configuring security features.							
4	1. Harden a Linux system. 2. Recovering passwords. 3. Online Malware Investigation Tools							

5	1. Packet Tracer: File and Data Integrity Checks, Explore File and Data Encryption. 2. Packet Tracer - Configure Wireless Router Hardening and Security. 3. Packet Tracer - Configure Access Control. 4. Configure Authentication and Authorization in Linux.
6	1. Packet Tracer - Configure Server-Based Authentication with TACACS+ and RADIUS. 2. Packet Tracer - Configure Named and Numbered Standard IPv4 ACLs. 3. Packet Tracer - Configure Extended ACLs. 4. Packet Tracer – Configure IPv6 ACLs.
7	1. Packet Tracer - Configure a Zone-Based Policy Firewall(ZPF). 2. Packet Tracer - Logging from Multiple Sources.
8	1. Snort and Firewall Rules. 2. Research Snort and CVE IDs, Classify alerts from Windows and Linux. 3. Packet Tracer - Use Diagnostic Commands.
9	1. Risk Management 2. Risk Analysis
10	1. Gather System Information After an Incident 2. Attack Analysis.
Additional Experiments	
1.	1. Develop Cybersecurity Policies and Procedures. (Assignment) 2. Create Your Personal Code of Ethical Conduct. (Assignment) 3. Recommend Security Measures to Meet Compliance Requirements (Assignment) 4. Review the features of an example of a penetrating testing vulnerability report. (Assignment) 5. Implementation of Security Controls (Assignment)
Reference Books:	
1	Cyber Security Essentials, Cisco Skills for All.
2	Cybersecurity Essentials, Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short, Sybex, O'Reilly Publications.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3209	PC	0	0	3	1.5	15	35	3Hrs.
IoT LAB								
(For CIC)								
Course Objectives: The main objective of the course is to provide students to								
1.	Know how to use Linux Commands and python programs using Raspberry pi module.							
2.	Know how to use Raspberry pi module for various IoT Applications.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Use sensors, actuators, Raspberry pi module in IoT applications.							K3
2.	Design and Develop various IoT applications.							K4
LIST OF PROGRAMS								
Following are some of the programs that a student should be able to write and test on an Raspberry Pi, but not limited to this only.								
1	Start Raspberry Pi and try various Linux commands in command terminal window: ls, cd, touch, mv, rm, man, mkdir, rmdir, tar, gzip, cat, more, less, ps, sudo, cron, chown, chgrp, ping etc.							
2	Run some python programs on Pi like: - Read your name and print Hello message with name - Read two numbers and print their sum, difference, product and division. - Word and character count of a given string - Area of a given shape (rectangle, triangle and circle) reading shape and appropriate values from standard input - Print a name 'n' times, where name and n are read from standard input, using for and while loops. - Handle Divided by Zero Exception. - Print current time for 10 times with an interval of 10 seconds. - Read a file line by line and print the word count of each line.							
3	Light an LED through Python program.							
4	Get input from two switches and switch on corresponding LEDs.							
5	Flash an LED at a given on time and off time cycle, where the two times are taken from a file.							
6	Flash an LED based on cron output (acts as an alarm).							
7	Switch on a relay at a given time using cron, where the relay's contact terminals are connected to a load.							
8	Get the status of a bulb at a remote place (on the LAN) through web. The student should have hands on experience in using various sensors like temperature, Humidity, smoke, light, etc. and should be able to use control web camera, network, and							

	Relays connected to the Pi.
Reference Books:	
1.	Internet of Things: Architecture, Design Principles and Applications, Rajkamal, McGraw Hill Higher Education. 2017.
2.	Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madhetti, Universities Press, 1st edition, 2014.
3.	Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley, 1st edition, 2014.
4.	Getting Started with the Internet of Things CunoPfister, Oreilly. 2011.
5.	Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly (SPD), 2014.
e-Resources:	
1.	https://www.coursera.org/specializations/iot
2.	https://www.coursera.org/learn/raspberry-pi-platform



Code	Category	L	T	P	C	I.M	E.M	Exam
B20CI3210	PC	0	0	3	1.5	15	35	3Hrs.
CRYPTOGRAPHY AND NETWORK SECURITY LAB								
(For CIC)								
Pre-requisites: C or Java Programming								
Course Objectives: The main objective of the course is to provide students								
1.	Learn Traditional cryptographic techniques.							
2.	Understand basic concepts of security services with symmetric and asymmetric algorithms.							
3.	Understand user and message authentication with MD-5, SHA-1 and Digital Signature algorithms							
Course Outcomes: At the end of the course students will be able to								
S No	Outcome							Knowledge Level
1.	Apply Substitution and Transposition Techniques							K3
2.	Use symmetric and asymmetric key algorithms for cryptography.							K3
3.	Use Hashing and Digital Signature algorithms to generate message digest.							K3
Note: All Programs can do either in C Language or Java Language								
Syllabus								
LIST OF PROGRAMS								
1	Write a program that contains a string (char pointer) with a value “Hello world”. The program should AND or and XOR each character in this string with 127 and display the result.							
2	Write a program to perform encryption and decryption using the following algorithms. 1. Caesar cipher 2. Play fair Cipher 3.Pure Transposition Cipher							
3	Write a program to implement the DES algorithm logic.							
4	Write a program to implement the AES algorithm logic.							
5	Write a program to implement RSA algorithm.							
6	Implement the Diffie-Hellman Key Exchange mechanism.							
7	Calculate the message digest of a text using the SHA-1 algorithm.							
8	Calculate the message digest of a text using the MD5 algorithm.							
9	Implement the Signature Scheme using Digital Signature Standard.							
Reference Books:								
1.	Modern Cryptography: Theory and Practice, by Wenbo Mao, Prentice Hall PTR							
2.	Cryptography: Theory and Practice by Douglas R. Stinson, CRC press.							

Code	Category	L	T	P	C	I.M	E.M	Exam
B20HS3203	SOC	1	--	2	2	--	50	3Hrs.
SOFT SKILLS								
(Common to CIC, CE, EEE, & ME)								
Course Objectives:								
1.	To familiarize students with soft skills and how they influence their professional growth.							
2.	To build/refine the professional qualities/skills necessary for a productive career and to instill Confidence through attitude building.							
Course Outcomes: At the end of course, Students will be able to								
S.No	Outcome							Knowledge Level
1	Apply soft skills in the workplace and build better personal and professional relationships making informed decisions.							K3
2	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.							K3
SYLLABUS								
1.	Introduction to Soft Skills, Significance of Inter & Intra-Personal Communication							
2.	SWOT Analysis, Creativity & Problem Solving							
3.	LSRW, JAM, Presentation Skills							
4.	Building a positive attitude, Leadership & Teamwork							
5.	Goal Setting – Guidelines for Goal Setting							
6.	Group Discussion: Essential guidelines							
7.	Telephone Etiquette, Telephonic Interview							
8.	Resume Preparation: Common resume blunders, tips for betterment, Resume Review							
9.	Employability Skills: Emotional Intelligence, Report Writing, Social Consciousness and Social Entrepreneurship, Stress Management.							
10.	Awareness about Industry, Companies, Importance of researching the prospective workplace, Knowing about Selection Process							
11.	Interview Skills: Types of Interviews, Mock Interview, Do's and Don'ts of Interview.							
Text Books:								
1.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.							
2.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi							
Reference Books:								

1.	The Art of Public Speaking by Dale Carnegie
2.	The Leader in You by Dale Carnegie
3.	Emotional Intelligence by Daniel Golman
4.	Stay Hungry Stay Foolish by Rashmi Bansal
5.	I have a Dream by Rashmi Bansal.

Additional Materials

1.	https://www.youtube.com/watch?v=LTnI7cmpDZI
2.	https://www.youtube.com/watch?v=ic5O2sxhH9M
3.	https://www.youtube.com/watch?v=4ZQkYSpmOdU
4.	https://www.youtube.com/watch?v=d8p-5WcXoRs
5.	https://www.youtube.com/watch?v=yZOar04g4zk&t=94s



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC3202	MC	--	--	3	--	--	--	--
COMPETITIVE PROGRAMMING								
(For CIC)								
Course Objectives:								
1	To learn about Stacks, Queues, Linked Lists and Templates.							
2	To learn different Searching and Sorting Algorithms.							
Course Outcomes: At the end of the course Students will be able to								
S.No	Outcome							Knowledge Level
1	Implement standard problems on Matrix.							K3
2	Solve various classical problems on Stacks, Queues and Linked Lists.							K3
3	Apply searching and sorting algorithms on different data structures.							K3
4	Solve the coding tasks with the help of STL.							K3
SYLLABUS								
UNIT-I	Standard Problems on Matrix Matrix forms and representations, Spiral filling and printing, Clockwise and anti-clock wise rotations, Zig-Zag form and Pattern coding.							
UNIT-II	Standard Problems on Linked List Finding and removing duplicates in Linked list, Multiple links, Construct a doubly linked list from 2D Matrix, List Sorting using bubble Sort, Modular node in Linked List, Rotate doubly linked list by n nodes.							
UNIT-III	Classical Problems on Stacks and Queues Implementing two stacks in an array, Design a stack with operations on middle element, String reversal using Stack, Queue implementation using stack, Queue reverse using queue, reversing first k elements of a stack.							
UNIT-IV	Searching and Sorting techniques sequential Vs. binary searching. Fibonacci Search, Exponential Search, Classical sorting techniques: Sort an array of strings using selection sort, Counting sort, Insertion sort on doubly linked list, Quick sort on single Linked List, Merge sort on Doubly linked List, 3-way Merge Sort.							
UNIT-V	Standard Library templates Introduction to C++ language, C++ Features, Working on STLs, Algorithms-Sorting and searching, Functions, Containers, Iterators.							

References:	
1.	Data Structures Using C by E. BalaguruSamy, First Edition, McGraw Hill Education, 2017 .
2.	The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd, Second Edition ,2015.
3.	Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg,3rd Edition, Thomson, 2013.
4.	Object Oriented Programming Using C++: Alok Kumar Jagadev , Amiya Kumar Rath, Satchidananda Dehuri,PHI,2010.
e-Resources:	
1.	https://www.geeksforgeeks.org/
2.	https://www.tutorialspoint.com/



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Code	Category	L	T	P	C	I.M	E.M	Exam
B20HS3204	HS	2	--	--	--	--	--	--
GENDER SENSITIZATION								
(Common to All Programmes)								
Course Objectives:								
1.	To develop students' sensibility with regard to issues of gender in contemporary India.							
2.	To provide a critical perspective on the socialization of men and women.							
3.	To introduce students to information about some key biological aspects of genders.							
4.	To help students reflect critically on gender violence and workplace security.							
5.	To expose students to more egalitarian interactions between men and women.							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the important issues relating to gender in contemporary India.							K2
2.	Get sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender.							K2
3.	Attain a finer grasp of how gender discrimination works in our society and how to counter it.							K2
4.	Acquire insight into the gendered division of labour and its relation to politics and economics.							K2
5.	Develop a sense of appreciation for both men and women in all walks of life.							K3
Estd. 1980AUTONOMOUS								
SYLLABUS								
UNIT-I	Understanding Gender and Related Concepts - Gender in Everyday Life Introduction: Conceptual Connotation – Sex and Gender – Basic Gender Concepts - Gendered Socialization – Gender Stereotypes –Exploring Attitudes towards Gender – Gender Roles & Relationships - Myths – Gender in Indian society – Early days – Later Vedic Period –Medieval and British Period – Independent India.							
UNIT-II	Introduction to Gender Justice- Notion and Significance Division and Valuation of Work – Housework- The Invisible Work - “My Mother doesn’t work,” - Offences against Women –Fact and Fiction - Status of Women in Society – Gender and Human Rights - Gender Equality – Gender Justice – Notion and Significance							
UNIT-III	International and Constitutional Perspectives on Gender Equality The International Bill of Rights, 1979 –Declaration on the Elimination of Violence against women 1993 –The Rights of Women –Beijing Platform for Action 1995 – Constitutional Guarantees – Fundamental Rights – Equality.							

UNIT-IV	Gender and Culture Gender and Film - Gender and Electronic Media – Gender and Advertisement – Gender and Popular Literature – Gender Issues - Gender-Sensitive Behaviour – Gender being Together as Equals.
UNIT-V	Gender Violence- Within and Beyond Violence – Gender Violence – Types of Gender Violence –Gender Violence in Indian Perspective – -Women Specific Legislations for the Elimination of Violence Within and Beyond.
Reference Books:	
1.	“Towards A World Of Equals: A Bilingual Textbook on Gender” by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas, and Susie Tharu, Published by Telugu Akademi (2015).
2.	Ferber, Holcomb & Wentling, Sex, Gender & Sexuality: The New Basics, Oxford Univ. Press 2008.
3.	Flavia Agnes, Sudhir Chandra, Monmayee Basu, Women and Law in India, Oxford Univ. Press 2004.
4.	Mamta Rao, Law Relating to Women and Children, Eastern Book Co, Lucknow.
5.	K.I. Vibhute, Criminal Law, Lexis Nexis, 12th Edn.
6.	N. Prabha Unnithan (ed.), Crime & Justice in India, Sage Pub., 2013.
7.	Ritu Gupta, Sexual Harassment at Workplace, Lexis Nexis, 2013.
8.	IGNOU: Gender Sensitization: Society, Culture and Change (2019) BGSE001, New Delhi IGNOU.
Web links:	
1.	https://nptel.ac.in/courses/110105080
2.	https://www.youtube.com/watch?v=2Xfp2eiTte0
3.	https://www.youtube.com/watch?v=-FCEBe5VNcA&t=41s
4.	https://www.youtube.com/watch?v=7n9IOH0NvyY
5.	https://www.youtube.com/watch?v=dpC2jGqu4G0
6.	https://www.youtube.com/watch?v=kcW4ABcY3zI&t=99s
7.	https://www.youtube.com/watch?v=dIXw1PbnWKM
8.	https://www.youtube.com/watch?v=9bayaZ18_po
9.	https://www.youtube.com/watch?v=ZbLq23cGFV4&t=1662s
10.	https://www.youtube.com/watch?v=61aYvb0Vo68
11.	https://www.youtube.com/watch?v=728H4Khf7Gk&t=1793s
12.	https://www.youtube.com/watch?v=y2Yk-rSZ7PI
13.	https://www.youtube.com/watch?v=wSqFvcjDpos
14.	https://www.youtube.com/watch?v=AljDd7nj9wE
15.	https://www.youtube.com/watch?v=MKPM0f2fOjM